

Josef Bisits

Biological, Earth and Environmental Sciences,
The University of New South Wales.

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Google scholar: [Josef I. Bisits](https://scholar.google.com/citations?user=Josef_I._Bisits)

EDUCATION

- **Doctor of Philosophy**, The University of New South Wales Sydney
School of Mathematics and Statistics 2022–2025
 - Supervisors: Associate Professor Jan Zika, Professor Trevor McDougall and Dr Taimoor Sohail
 - Thesis: [Non-linear controls on ocean circulation and mixing in the high-latitude oceans](#)
 - Published version of PhD thesis [abstract](#)
- **Bachelor of Science (Advanced mathematics, honours)**, The University of New South Wales Sydney
School of Mathematics and Statistics 2016–2021
 - Major: Applied mathematics
 - Supervisors: Associate Professor Jan Zika and Dr Geoff Stanley
 - Thesis: Passive tracer mixing along isopycnal surfaces in turbulent flows
 - Honours grade: class 1
- **Bachelor of Music**, The University of Sydney Sydney
Sydney Conservatorium of Music 2007–2010
 - Major: Performance, double bass
 - Double bass instructor: Alex Henery

PUBLICATIONS

- Bisits, Josef I.** et al. “A simple formula to infer rates of numerical mixing in finite volume ocean models part two: application”. In prep.
- Zika, Jan D., **Bisits, Josef I.** et al. “A simple formula to infer rates of numerical mixing in finite volume ocean models part one: theory and example”. In prep.
- Bisits, Josef I.**, McDougall, Trevor J., and Zika, Jan D. Aug. 2026. “Determining how nonlinearities in the equation of state affect diffusive interfaces in the ocean”. In: *Journal of Fluid Mechanics* 1040, A33. DOI: [10.1017/jfm.2026.11889](https://doi.org/10.1017/jfm.2026.11889).
- Lloyd, Simon D., **Bisits, Josef I.** et al. 2026c. “Swimming in urban estuaries: understanding stormwater contamination events and recovery from historical data”. In: *Water Research*, p. 125686. DOI: [10.1016/j.watres.2026.125686](https://doi.org/10.1016/j.watres.2026.125686).
- Bisits, Josef I.**, Zika, Jan D., and Sohail, Taimoor. May 2025. “Cabbeling as a catalyst and driver of turbulent mixing”. In: *Journal of Fluid Mechanics* 1011, A17. DOI: [10.1017/jfm.2025.349](https://doi.org/10.1017/jfm.2025.349).
- Bisits, Josef I.**, Zika, Jan D., and Evans, Dafydd Gwyn. Dec. 2024. “Does Cabbeling Shape the Thermohaline Structure of High-Latitude Oceans?” In: *Journal of Physical Oceanography* 54 (12), pp. 2419–2430. DOI: [10.1175/JPO-D-24-0061.1](https://doi.org/10.1175/JPO-D-24-0061.1).
- Bisits, Josef I.**, Stanley, Geoffrey J., and Zika, Jan D. Feb. 2023. “Can We Accurately Quantify a Lateral Diffusivity from a Single Tracer Release?” In: *Journal of Physical Oceanography* 53 (2), pp. 647–659. DOI: [10.1175/JPO-D-22-0145.1](https://doi.org/10.1175/JPO-D-22-0145.1).

SELECTED CONFERENCE PROCEEDINGS AND SEMINARS

ORAL PRESENTATIONS

- Bisits, Josef I.**, Zika, Jan D., and Sohail, Taimoor. Feb. 2026. “Cabbeling as a catalyst and driver of turbulent mixing”. In: *Ocean Sciences*.
- Bisits, Josef I.** Oct. 2025. “Non-linear controls on ocean circulation and mixing in the high-latitude oceans”. In: *Australis Centre for Excellence in Antarctic Science seminar series*.

- Bisits, Josef I.**, Zika, Jan D., and Evans, Daffyd Gwynn. June 2024. “Does cabbeling shape the thermohaline structure of high-latitude oceans?” In: *Gordon Research Seminar on Ocean Mixing*.
- Bisits, Josef I.**, Zika, Jan D., and Evans, Daffyd Gwynn. Feb. 2024. “Does cabbeling shape the thermohaline structure of high-latitude oceans?” In: *Australian Meteorological and Oceanographic Society annual conference*.
- Bisits, Josef I.**, Stanley, Geoffrey J., and Zika, Jan D. July 2023. “Can we accurately quantify a lateral diffusivity from a single tracer release?” In: *XVIII General Assembly of the International Union of Geodesy and Geophysics (IUGG)*.
- Bisits, Josef I.**, Stanley, Geoffrey J., and Zika, Jan D. Nov. 2022. “Can we accurately quantify a lateral diffusivity from a single tracer release?” In: *Australian Meteorological and Oceanographic Society annual conference*.

POSTER PRESENTATIONS

- Bisits, Josef I.**, Zika, Jan D., and Sohail. June 2024. “The effect of non-linear processes on mixing”. In: *Gordon Research Conference on Ocean Mixing*.
- Bisits, Josef I.**, Zika, Jan D., and Sohail, Taimoor. Nov. 2024. “The effects of cabbeling on mixing and energetics in polar oceans”. In: *Australian Antarctic Research Conference*.
- Bisits, Josef I.**, Zika, Jan D., and Evans, Daffyd Gwynn. July 2023. “Does cabbeling shape the thermohaline structure of high-latitude oceans?” In: *Physics of the Ocean Summer School*.
- Bisits, Josef I.**, Stanley, Geoffrey J., and Zika, Jan D. Nov. 2022. “Can we accurately quantify a lateral diffusivity from a single tracer release?” In: *Australian Centre for Excellence in Antarctic Science workshop*.

SCIENTIFIC EXPERIENCE

RESEARCH

- **Postdoctoral research associate**, Biological Earth and Environmental Sciences UNSW, Sydney
Diagnosing numerical mixing in finite volume models of fluid flow with A/Prof. Jan D. Zika. 2025-current
- **Research assistant**, School of Mathematics and Statistics UNSW, Sydney
Numerical modeling and analysis of tracer release experiments in turbulent flows with A/Prof. Jan Zika. 2022
- **Research assistant**, Faculty of Built Environment UNSW, Sydney
Historical water quality data analysis to develop a predictive water quality model with A/Lec. Simon Lloyd. 2022

INDUSTRY

- **Ocean modelling contractor** Submarine
Build and validate a model to simulate tracer release experiments in Port Angeles Harbour, Washington, USA. 2025-2026

TEACHING

MATHEMATICS

- **Tutoring** UNSW, Sydney
I have been involved in mathematics tutoring and marking since my honours year at UNSW including: 2021–Current
 - Mathematics Drop in Centre;
 - Mathematics 1A (MATH1131/1141);
 - Mathematics 1B (MATH1231/1241);
 - Numerical Methods and Statistics (MATH2089);
 - Introduction to Atmosphere and Ocean Dynamics (MATH2241);
 - Fluids, Oceans, and Climate (MATH3261-5285); and
 - Third year project (MATH3001) marker.

MUSIC

- **Private and ensemble double bass tuition** Sydney
During my double bass career I tutored private students and youth orchestra ensembles including: 2013–2022

- Sydney Youth Orchestras ensembles;
- Penrith Youth Orchestra (PYO), and PYO academy (as a member of the Australian Chamber Orchestra);
- Playerlink (as a member of the Sydney Symphony Orchestra);
- Private students attended the Australian Institute of Music and successfully auditioned for Australain Youth Orchestra programs.

PROGRAMMING SKILLS

I have a wide experience in programming which includes:

- using High Performance Computing systems to run models on both CPU's and GPU's;
- building models for ocean and fluid dynamics simulations; and
- data visualisation.

My programming language of choice is **julia**. Some packages I am involved with are:

- Author `StaircaseShenanigans.jl`, `TwoLayerDirectNumericalShenanigans.jl`, `RasterHistograms.jl`;
- Developer `PassiveTracerFlows.jl`; and
- Contributor `FourierFlows.jl`, `GeophysicalFlows.jl`, `Oceananigans.jl` amongst others.

I am also fluent with , , , , $\text{\LaTeX}$ and `git`.

PROFESSIONAL ACTIVITIES

- **Peer reviewer** 2023-current
 - Carried out peer reviews of articles submitted to:
 - Journal of Geophysical Research: Oceans;
 - EGUSphere;
 - Ocean Modelling.
- **Conference session chair** UNSW, Sydney
2024
 - School of Mathematics and Statistics postgraduate conference.

ORCHESTRAL EXPERIENCE

PROFESSIONAL

- **Sydney Symphony Orchestra** Sydney
 - Regular casual double bassist, full time contract musician in 2014, 2015 and September 2017–September 2018. 2009–current
- **Australian Chamber Orchestra** Sydney
 - Regular casual double bassist. 2013–current
- **Opera Australia Orchestra** Sydney
 - Regular casual double bassist. 2016–current
- **Click Track Music** Sydney
 - Regular casual double bassist. 2009–current
- **Tasmanian Symphony Orchestra** Tasmania
 - Casual double bassist. 2013–2018
- **Auckland Philharmonia** Auckland
 - Casual double bassist. 2013
- **Ulster Orchestra** Belfast
 - Held a trial for double bass section leader. Winter 2012

TRAINING PROGRAMS

- **Emerging Artist**, Australian Chamber Orchestra Sydney
Mentor: Maxime Bibeau. 2013
- **Southbank Sinfonia** London
Principal double bassist, twice appeared as a soloist. 2011-2012
- **Sydney Symphony Fellowship**, Sydney Symphony Orchestra Sydney
Also appeared with Sydney Sinfonia from 2008-2010. 2010
- **Australian Youth Orchestra** Australia
Double bass section player. 2007–2008
- **Sydney Youth Orchestra** Sydney
Principal double bassist, appeared as double bass soloist in 2006. 2006–2008

SCHOLARSHIPS AND AWARDS

- Best applied mathematics talk at UNSW School of Mathematics and Statistics postgraduate conference. 2024
- Michael Tallis Research travel award. 2023
- Australian Government Research Training Program Scholarship. 2022–2025
- School of Mathematics and Statistics research scholarship top-up. 2022–2025
- Australian Centre for Excellence in Antarctic Research PhD student. 2022–2025
- Michael Bannigan scholarship for academic achievement. 2010
- Corina d'Hage string scholarship. 2008–2010
- Winner of Sydney Youth Orchestra concerto competition. 2006

SELECTED OUTREACH ACTIVITIES

- **Helper for Girls do the Maths!**, *University of New South Wales*. 2022, 2023
- **Helper for Experience UNSW Science day**, *University of New South Wales*. 2022
- **Climate @ maths display for UNSW open day**, *University of New South Wales*. 2022, 2023, 2024
- **Helper for uDASH work experience workshop**, *University of New South Wales*. 2022

REFERENCES

- **Associate Professor Jan Zika**, *University of New South Wales, Sydney*. Email: j.zika@unsw.edu.au
- **Dr Taimoor Sohail**, *Research Fellow, University of Melbourne*. Email: taimoor.sohail@unimelb.edu.au
- **Emeritus Professor Trevor McDougall**, *University of New South Wales, Sydney*. Email: trevor.mcdougall@unsw.edu.au
- **Dr Geoff Stanley**, *Banting Postdoctoral Fellow, University of Victoria, Canada*. Email: gstanley@uvic.ca
- **Associate Lecturer Simon Lloyd**, *University of New South Wales, Sydney*. Email: s.lloyd@unsw.edu.au
- **Benjamin Ward**, *Tutti double bass, Sydney Symphony Orchestra*. Email: 2benward@gmail.com
- **Edmund Bastian**, *Tutti double bass, Opera Australia Orchestra*. Email: tanya.ed@bigpond.com